



CASSINI #EUSpace
Hackathons & Mentoring

Big Ideas Campaign Session 1

From real challenges to
impactful solutions

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Session Overview

In this session we will:

-  Explain how the challenges were defined
-  Understand the focus of the hackathon
-  Present the key challenge areas
-  Clarify expectations for participants



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Space for Water: why this hackathon?

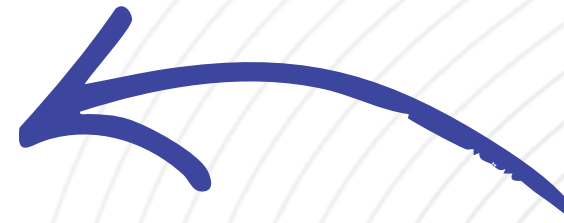
This hackathon focuses on applying space technologies to real challenges in water management.

We are addressing

- Water scarcity and resource management
- Infrastructure efficiency
- Water quality and contamination
- Climate-related risks (droughts & floods)

Using

- Satellite data (Copernicus & Galileo)
- Data analysis and modelling
- Decision-support tools



From focus group to challenge areas

Who was involved

- Water management experts
- Data and technology specialists
- Space and satellite applications pros



From focus group to challenge areas

What we did

- Identified real needs from the territory
- Contrasted different perspectives
- Prioritized high-impact areas

Result:

Challenges grounded in reality, not assumptions.



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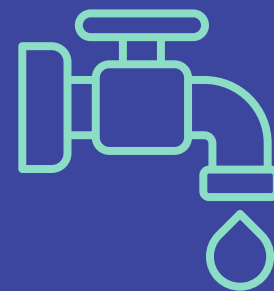


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The challenge framework



CHALLENGE 1
Water loss reduction



CHALLENGE 2
Water consumption analysis



CHALLENGE 3
Water quality monitoring



CHALLENGE 4
Coastal erosion & pollution impact



CHALLENGE 5
Urban flood prediction & hydrological risk



CHALLENGE 6
Snow monitoring & Water availability

CHALLENGE 1

Water loss reduction



Detect leaks in urban water distribution systems



Combine satellite data with IoT, consumption and pressure data



Improve monitoring and localization of incidents



Opportunity

Combine satellite data with analytics to enhance visibility, reduce losses and optimize infrastructure management



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CHALLENGE 2

Water consumption analysis

-  Analyze water use across sectors (agriculture, urban, industry)
-  Understand the relationship between consumption and resource availability
-  Track activities and infrastructure using geolocation



Opportunity

Enable better planning, identify inefficiencies and anticipate water scarcity scenarios

CHALLENGE 3

Water quality monitoring

-  Monitor rivers, reservoirs and coastal areas
-  Detect contamination events (e.g. cyanobacteria)
-  Improve early warning and response systems



Opportunity

Enable early risk detection, protect public health and improve ecosystem conservation



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CHALLENGE 4

Urban flood prediction & hydrological risk



Identify flood-prone areas



Estimate flood extent during extreme events



Analyze factors affecting water retention (including vegetation)



Opportunity

Improve prediction models, early warning systems and emergency response capabilities



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CHALLENGE 5

Coastal erosion & pollution impact



Monitor coastline evolution and erosion



Identify vulnerable areas (e.g. deltas and coastal zones)



Analyze impact of human activity (ports, urban pressure, pollution)



Opportunity

Support environmental protection, improve coastal management and enable climate adaptation strategies



CHALLENGE 6

Snow monitoring & water availability



Analyze snow coverage and its evolution



Anticipate water availability in river basins



Support hydrological and reservoir management



Opportunity

Improve water resource prediction, reservoir management and drought preparedness



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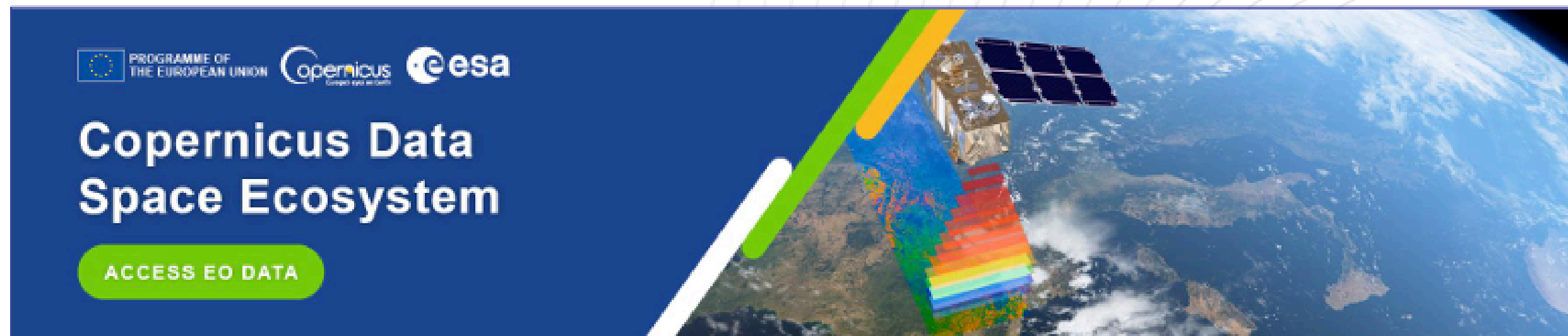
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The Copernicus Data Space Ecosystem

- We are working with the [Copernicus Data Space Ecosystem](#) to give you free access to cloud processing infrastructure and data storage for the duration of the event.
- The package includes:
 - **Processing tools:** Access APIs, versatile tools and web-based environments
 - **Data access:** Data & information at your fingertips
 - **Training & support:** 2 Training sessions and technical support (4h) during the Hackathon Weekend



Local Data and Tools



- Platforms for viewing and downloading Sentinel-2 satellite imagery (RGB and infrared).
- High-resolution data and topographic layers (VISSIR).
- Ground movement information based on Sentinel-1 radar.
- Up-to-date land cover maps.
- Tools for creating and publishing your own maps (Instamaps).
- Historical comparators for analyzing land evolution.

OPEN COSMOS

- Data from hyperspectral satellites
- Access to other types of satellite data as needed by the teams



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Q & A

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